Homework 2

BSTA 551

Directions

Please turn in this homework on Sakai. Please submit your homework in pdf format or in html rendered from a qmd (or Rmd) file (please also include source qmd or Rmd). You can type your work on your computer or submit a single file with photos of your written work or any other method that can be turned into a pdf.

You must show all of your work to receive credit.

This homework covers material from Lessons 3, 4, and 5: MVUE and the Cramér-Rao Lower Bound, Method of Moments and Maximum Likelihood Estimation, and Sufficient Statistics. Show all work for full credit. For problems requiring R, include your code and output.

For simulation problems, set your seed to `set.seed(551)` for reproducibility.

For the qmd of this Homework page, see this github link

Questions

Problem 1: Fisher Information and CRLB

A clinical trial is testing a new treatment for hypertension. The time (in days) until a patient achieves target blood pressure is modeled as an Exponential distribution with rate parameter λ (where $E(X) = 1/\lambda$).

In a pilot study, the following times (in days) were observed for $n = 8$ patients:

$$X = \{12, 8, 15, 6, 10, 14, 9, 11\}$$

(a) Calculate the Fisher Information $I(\lambda)$ for a single observation from an Exponential(λ) distribution. Show your derivation using the definition $I(\lambda) = E[-\ell''(\lambda)]$.

(b) State the Cramér-Rao Lower Bound (CRLB) for unbiased estimators of λ based on a sample of size n.

(c) The MLE of λ is $\hat{\lambda} = 1/\bar{X}$. This estimator is biased. Find an unbiased estimator of λ and compute its variance. Does this unbiased estimator achieve the CRLB?

(d) Using R, compute the MLE $\hat{\lambda}$ from the pilot study data. Also compute the CRLB at this estimated value. Interpret what this bound tells us about estimation precision.

```r
# Pilot study data
bp_times <- c(12, 8, 15, 6, 10, 14, 9, 11)

# Your code here
```

Problem 2: Method of Moments (Shifted Exponential Distribution)

The time (in hours) from hospital admission to discharge for patients undergoing a minor outpatient procedure can be modeled with a shifted (two-parameter) Exponential distribution. This accounts for a minimum required observation time θ plus additional random waiting time. The pdf is:

$$f(x; \lambda, \theta) = \lambda e^{-\lambda(x-\theta)}, \quad x \geq \theta, \quad \lambda > 0, \quad \theta > 0$$

For this distribution:

$$E(X) = \theta + \frac{1}{\lambda}, \quad \mathrm{Var}(X) = \frac{1}{\lambda^2}$$

(a) Derive the Method of Moments estimators $\tilde{\lambda}$ and $\tilde{\theta}$ in terms of the sample mean $\bar{X}$ and sample standard deviation S.

(b) The following discharge times (in hours) were recorded for $n = 10$ patients:

$$3.2,\ 4.1,\ 2.8,\ 5.3,\ 3.5,\ 4.8,\ 3.1,\ 6.2,\ 4.4,\ 3.9$$

Using R, calculate the MME estimates $\tilde{\lambda}$ and $\tilde{\theta}$. Verify that your estimate of θ is less than the minimum observed value (a necessary condition for validity).

```r
# Discharge times (hours)
discharge_times <- c(3.2, 4.1, 2.8, 5.3, 3.5, 4.8, 3.1, 6.2, 4.4, 3.9)

# Your code here
```

Problem 3: Maximum Likelihood Estimation (Devore 7.25)

Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with pdf:

$$f(x; \theta) = (\theta + 1)x^{\theta}, \quad 0 \le x \le 1, \quad \theta > -1$$

(a) Derive the maximum likelihood estimator $\hat{\theta}$ of θ.

(b) For a sample of size $n = 5$ with observations $x_1 = 0.92, x_2 = 0.79, x_3 = 0.85, x_4 = 0.95, x_5 = 0.88$, compute the MLE $\hat{\theta}$.

(c) Use the invariance property of MLEs to find the MLE of $\tau(\theta) = \frac{\theta+1}{\theta+2}$, which represents $E(X)$ for this distribution.

Problem 4: Sufficient Statistics (Medical Application)

A researcher is studying the number of adverse events reported per patient in a drug trial. The number of adverse events X follows a Poisson(μ) distribution.

(a) For a random sample $X_1, \dots, X_n$ from Poisson(μ), use the Neyman Factorization Theorem to show that $T = \sum_{i=1}^{n} X_i$ is a sufficient statistic for μ.

(b) Show that the MLE $\hat{\mu} = \bar{X}$ is a function of the sufficient statistic T.

(c) In a clinical trial with $n = 50$ patients, suppose the following summary statistics are recorded:

$$\sum_{i=1}^{50} X_i = 72$$

A colleague suggests using a different estimator: $\tilde{\mu} = \frac{\text{number of patients with } X_i > 0}{n}$.

Using R simulation, generate 1000 samples from Poisson($\mu = 1.5$) with $n = 50$ and compare the bias, variance and MSE of $\hat{\mu}$ and $\tilde{\mu}$.

```r
# Your simulation code here
n <- 50
mu_true <- 1.5
n_sims <- 1000

# Hint: For each simulation:
# 1. Generate n Poisson(mu) observations
# 2. Compute mu_hat = mean(X)  (MLE, based on sufficient statistic)
# 3. Compute mu_tilde = mean(X > 0)  (proportion with events)
# 4. Compare MSEs
```

Problem 5: MLE for Normal Variance (Devore 7.31 Modified)

Measurement errors in a laboratory instrument are assumed to follow a Normal distribution with known mean $\mu = 0$ and unknown variance σ^2. A sample of n measurements yields observations $X_1, \dots, X_n$.

(a) Derive the MLE of σ^2.

(b) Is this MLE unbiased? If not, find the bias.

(c) A technician takes $n = 15$ measurements and records:

$$\sum_{i=1}^{15} X_i^2 = 4.8$$

Calculate the MLE of σ^2 and provide an unbiased estimate.